LOCAL STRUCTURE OF A FINITE TYPE MORPHISM AT A POINT OF ITS FIBRE

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 13407-13429, immediately following the statement at 13382-13405.

Editorial note. Two cross-chapter pointers (Algebra, Morphisms) plus one same-chapter lemma are recorded in assembly.external_references.

Editorial note. The statement's four numbered conclusions are all checked inside the span; the closing sentence handles the permanence assertion.

Allowed standard prerequisites (editorial).

- Refined structure of a quasi-finite morphism over a polynomial algebra. Algebra, lemma algebra-lemma-refined-quasi-finite-over-polynomial-algebra, cited in the proof. Produces the morphism to A^n_S with purely transcendental residue field extension that the local model needs.
- Making quasi-finiteness finite after an elementary etale neighbourhood. chapter lemma lemma-etale-makes-quasi-finite-finite-at-point, cited at proof line 13420. Provides the elementary etale neighbourhood (Y, y) -> (A^n_S, pi(x)) and the open X' of the base change over which the morphism becomes finite.
- Permanence of finite presentation. Morphisms, lemma morphisms-lemma-finite-presentation-permanence, cited in the proof. Keeps the finite presentation hypothesis along the constructed diagram.

Source reference. Original reference 0520: focused-seeds/geometry/S9-morph-local-structure-finite-type/ref-tags.txt, source lines 4–4. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

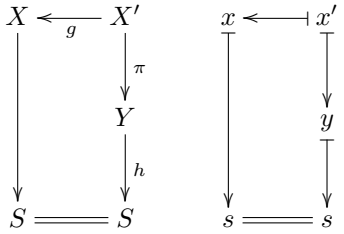
Source reference. Original lemma 02LK: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 11924–11924. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 02FV: focused-seeds/geometry/S9-morph-local-structure-finite-type/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $f : X \to S$ be a morphism. Let $x \in X$ and set $s = f(x)$. Assume that f is locally of finite type and that $n = \dim_x(X_s)$. Then there exists a*

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

commutative diagram



and a point $x' \in X'$ with $g(x') = x$ such that with $y = \pi(x')$ we have

(1) *$h : Y \to S$ is smooth of relative dimension n,*
(2) *$g : (X', x') \to (X, x)$ is an elementary étale neighbourhood,*
(3) *π is finite, and $\pi^{-1}(\{y\}) = \{x'\}$, and*
(4) *$\kappa(y)$ is a purely transcendental extension of $\kappa(s)$.*

Moreover, if f is locally of finite presentation then π is of finite presentation.

Proof. The problem is local on X and S, hence we may assume that X and S are affine. By Algebra, Lemma 0520 after replacing X by a standard open neighbourhood of x in X we may assume there is a factorization

$$X \xrightarrow{\pi} \mathbf{A}^n_S \longrightarrow S$$

such that π is quasi-finite and such that $\kappa(\pi(x))$ is purely transcendental over $\kappa(s)$. By Lemma 02LK there exists an elementary étale neighbourhood

$$(Y, y) \to (\mathbf{A}^n_S, \pi(x))$$

and an open $X' \subset X \times_{\mathbf{A}^n_S} Y$ which contains a unique point x' lying over y such that $X' \to Y$ is finite. This proves (1) – (4) hold. For the final assertion, use Morphisms, Lemma 02FV. □